LEARNER GUIDE

Solving by substitution

Equations, Inequalities and Simultaneous Relationships · Lesson 7. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

Reading a crossing off a graph is quick but only as accurate as the drawing. **Substitution** finds it exactly. If one equation says $y = 2x + 1$, then y and $2x + 1$ are the *same number*, so wherever the other equation has a y you may put $(2x + 1)$ in its place. That leaves a single equation in one unknown — solve it for x, put x back to get y, and **check in both** equations. If neither equation is in the form $y = \ldots$, rearrange one first.

What you are learning to notice

- If y equals an expression, you may substitute that expression wherever y appears.
- Substituting turns two equations in two unknowns into **one** equation in one unknown.
- Solve for x, then **back-substitute** to find y.
- Check the pair in *both* original equations — both must be true.

Moving through it, one step at a time

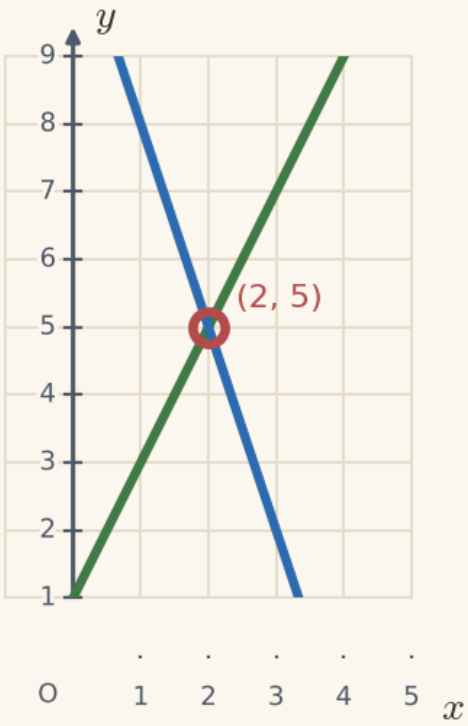
Pick the equation already in the form $y = \ldots$ (or rearrange one so it is). Substitute that expression for y in the other equation. Expand any bracket and collect like terms, giving a single equation in x; solve it. Put your x back into $y = \ldots$ to find y. Finally substitute the pair (x, y) into both original equations to check they are each satisfied.

Worked examples

Two full examples to follow. Read them slowly — the aim is to see how each line follows from the one before, not to memorise a rule. It can help to cover the steps and predict the next line before you reveal it.

Example 1 — Substitute, solve, check

Solve $y = 2x + 1$ and $3x + y = 11$.

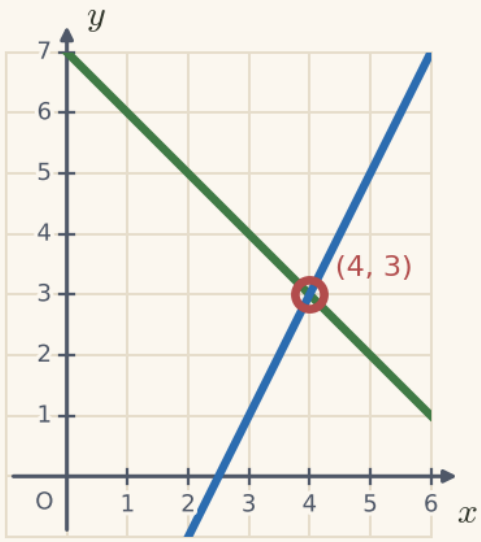


1. Substitute $(2x + 1)$ for y in the second equation.	$3x + (2x + 1) = 11$
2. Expand and collect.	$5x + 1 = 11$
3. Solve for x.	$x = 2$
4. Back-substitute for y, then check both.	$y = 5$; and $3\times2 + 5 = 11$ ✓

(2, 5) — substituting gives $5x + 1 = 11$, so $x = 2$ and $y = 5$; it satisfies both. This is the crossing.

Example 2 — Rearrange first

Solve $x + y = 7$ and $2x - y = 5$ — neither is in $y = \ldots$ form yet.



1. Rearrange the first to make y the subject.	$y = 7 - x$
2. Substitute into the second.	$2x - (7 - x) = 5$
3. Expand and collect, then solve.	$3x - 7 = 5$, so $x = 4$
4. Back-substitute and check.	$y = 3$; and $4 + 3 = 7$ ✓

(4, 3) — rearranging to $y = 7 - x$ lets substitution run as usual; $x = 4$, $y = 3$ satisfies both.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (substituting a value or expression, rearranging to make a variable the subject, and solving a two-step equation), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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